

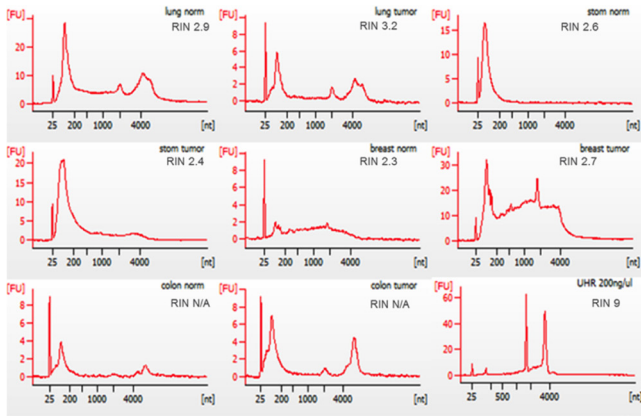


Figure 2: RNA Quality from FFPE Samples— RNA isolated from FFPE samples was examined using an Agilent Bioanalyzer. RNA Integrity Numbers (RINs)⁵ were calculated from the Bioanalyzer traces.

Evaluating RNA Quality

Many researchers use the Agilent RNA Integrity Number (RIN)⁵ to determine RNA quality for gene expression analysis (Figure 2). However, we have found that RIN values from degraded FFPE samples are not a sensitive measure of RNA quality. They are also not a reliable predictor of successful library preparation (Table 1 and Figure 3).

Because mean RNA fragment size is a more reliable determinant of RNA quality for the TruSeq RNA Access Library Preparation Kit (Figure 3), Illumina developed the DV₂₀₀ metric—the percentage of RNA fragments > 200 nucleotides. Using DV₂₀₀ to assess FFPE RNA quality enables accurate fine-tuning of the minimal RNA input required for successful library preparation. By adjusting RNA input amounts, high-quality libraries can be prepared from poor-quality FFPE samples (Table 2). The recommended quantities of starting material correspond to a range of DV₂₀₀ values; note that using RNA samples with DV₂₀₀ < 30% is not recommended (table 2). Using the DV₂₀₀ metric to determine the RNA input requirement also ensures robust, reproducible results. Excellent correlation (R²=0.99) was observed between the normalized gene expression counts from 3 different FFPE tumor samples of varied RNA quality (Figure 4).

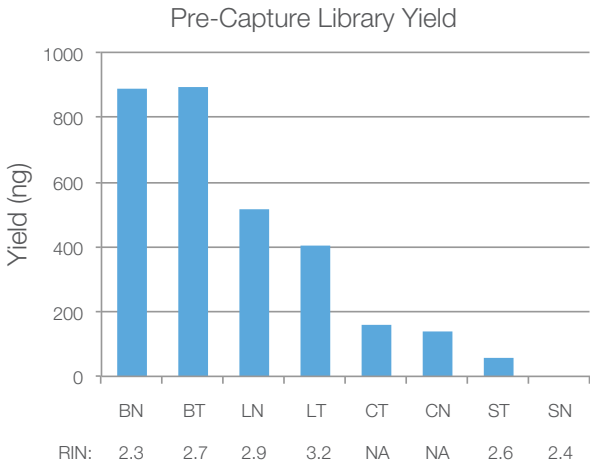
The DV₂₀₀ value can be easily calculated from Fragment Analyzer or Bioanalyzer traces (Figure 5). Customized Illumina FFPE RNA DV₂₀₀ Fragment Analyzer methods for standard sensitivity and high sensitivity RNA sample kits can be downloaded from the Advanced Analytical website at www.aati-us.com/product/fragment-analyzer.

Table 1: RIN and DV₂₀₀ Values From FFPE Samples

Sample	RIN	DV ₂₀₀ ^a
Breast Normal	2.3	77
Breast Tumor	2.7	71
Lung Normal	2.9	55
Lung Tumor	3.2	50
Colon Normal	N/A	32
Colon Tumor	N/A	39
Stomach Tumor	2.4	30
Stomach Normal	2.6	8

^aThe “DV₂₀₀” is the percentage of RNA fragments > 200 nucleotides. Although RIN values for these samples lie within a relatively narrow range (2.3–3.2), the size distribution of the RNA varies greatly among the samples.

A



B

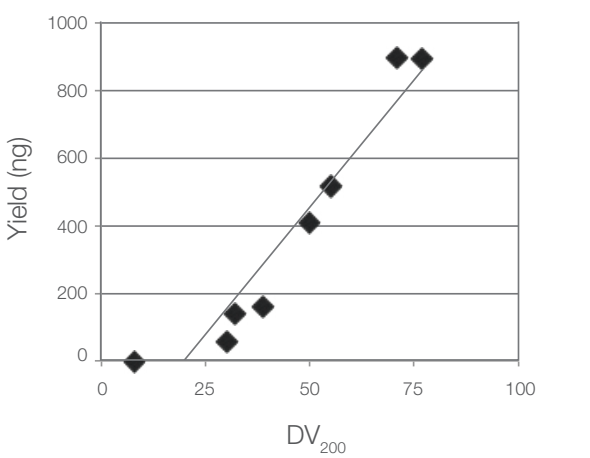


Figure 3: RIN versus DV₂₀₀ and Library Yield— (A) Libraries were prepared from 20 ng total RNA from 8 FFPE samples. RIN values range from 2.3 – 3.2 and do not show a correlation to precapture library yield. (B) High correlation (R² = 0.91) between precapture library yield and DV₂₀₀.

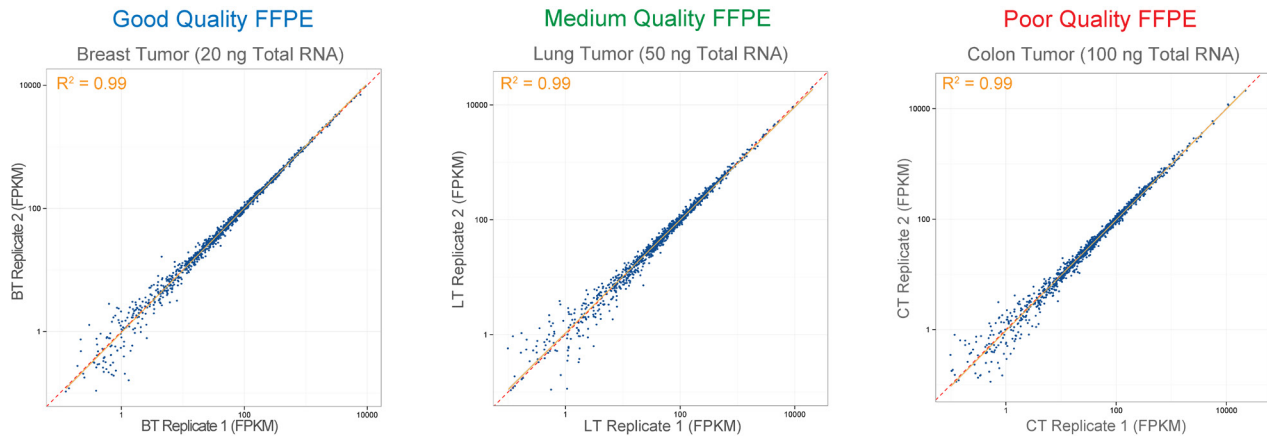


Figure 4: High concordance of gene expression between replicates across FFPE samples of varied quality—TruSight RNA Pan-Cancer libraries were prepared from High, Medium, and Low Quality FFPE from 20 ng, 50 ng, and 100 ng of total RNA, respectively. Normalized gene expression counts, in fragments per kilobase per million mapped reads (FPKM), were calculated by BaseSpace® TopHat Alignment App.⁴

Table 2: Recommended RNA Input Based on DV₂₀₀

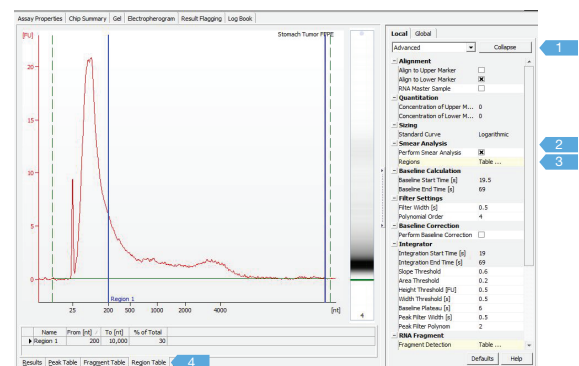
Parameter Quality*†‡	DV ₂₀₀	Recommended Input Quantity	
		TruSeq RNA Access	TruSight RNA Pan-Cancer or TruSight RNA Fusion
High	> 70%	20 ng	20 ng
Medium	50–70%	20–40 ng	20–50 ng
Low	30–50%	40–100 ng	50–100 ng
Too Degraded	< 30%	Not Recommended	

* to ensure successful library prep, we recommend using an RNA isolation method that includes a reverse-crosslinking step and DNase1 treatment (we used QIAGEN RNeasy FFPE Kit or QIAGEN AllPrep DNA/RNA FFPE Kit).
† FFPE RNA concentration was determined by NanoDrop.
‡ To ensure best performance on samples close to edge of a quality classification, err towards the higher end of the input recommendation.

Summary

Next-generation sequencing approaches applied to FFPE preserved tissue samples along with their associated clinical data offer an invaluable resource for translational research. The RNA enrichment library prep kits allow researchers to utilize FFPE and other challenging samples in their NGS studies. For these studies, the DV₂₀₀ of RNA fragments, as determined by a Fragment Analyzer or Bioanalyzer trace, is a reliable predictor for successful RNA sequencing results using the TruSeq RNA Access Library Preparation Kit, or the TruSight RNA Pan-Cancer Panel.

A



B

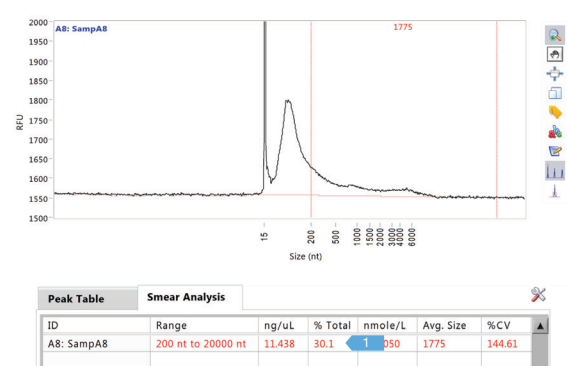


Figure 5: Calculate DV₂₀₀—(A) The percentage of RNA fragments > 200 nt (DV₂₀₀) can be calculated from a Bioanalyzer trace by performing a Smear Analysis as follows: 1) Under the *Local* tab, change *Normal* to *Advanced*. 2) Check box for *Smear Analysis*. 3) Click *Table*, add a region, and enter 200–10,000 bp in the popup window. 4) Select the *Region Table* tab in the trace window to display the results. (B) The Fragment Analyzer system has a streamlined solution for DV₂₀₀ analysis. ProSize™ software automatically configures the > 200 nt smear analysis parameters, and the DV₂₀₀ result is displayed as the % Total value within the data table.

Ordering Information

Product	Catalog No.
TruSight RNA Pan-Cancer Panel Set A (48 samples, 12 indexes)	RS-303-1002
TruSight RNA Pan-Cancer Panel Set B (48 samples, 12 indexes)	RS-303-1003
TruSight RNA Fusion Panel Set A (48 samples, 12 indexes)	RS-304-1002
TruSight RNA Fusion Panel Set B (48 samples, 12 indexes)	RS-304-1003
TruSeq RNA Access Library Prep Kit Set A (48 samples, 12 indexes)	RS-301-2001
TruSeq RNA Access Library Prep Kit Set B (48 samples, 12 indexes)	RS-301-2002

Learn More

For more FFPE RNA solutions, visit

www.illumina.com/applications/sequencing/rna/low-quality-ffpe-rna-seq.ilmn

To learn more about Illumina RNA enrichment assays, visit

TruSight RNA Pan-Cancer: www.illumina.com/RNAPanCancer

TruSight RNA Fusion: www.illumina.com/RNAFusion

TruSeq RNA Access: www.illumina.com/products/truseq-rna-access-kit.ilmn

References

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2. Penland SK, Keku TO, Torrice C, He X, Krishnamurthy J, Hoadley KA, et al. RNA expression analysis of formalin-fixed paraffin-embedded tumors. *Lab Invest* 2007;794: 383–391.
3. Norton N, Sun Z, Asmann YW, Serie DJ, Necela BM, et al. Gene expression, single nucleotide variant and fusion transcript discovery in archival material from breast tumors. *PLOS One* 2013;8(11): e81925.
4. Illumina. BaseSpace Core Applications: TopHat Alignment App. support. illumina.com/sequencing/sequencing_software/basespace/documentation.ilmn. Accessed May 25, 2016.
5. Agilent Technologies. (2004) RNA Integrity Number (RIN) - Standardization of RNA Quality Control Publication PN 5989-1165EN. www.chem.agilent.com. Accessed May 25, 2016.

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